LIMITS

A limit is the value that a function approaches as the value of x gets closer and closer to a particular number.

Let us understand with an example.

Consider the function:

$$f(x) = x + 2$$

Suppose we want to know what happens when $x$approaches 3.

x	$f(x) = x + 2$
2	4
2.5	4.5
2.9	4.9
2.99	4.99
3	5
3.01	5.01
3.1	5.1

We can observe that when x approaches 3, the value of the function approaches 5.

Therefore,

$$\boxed{\lim_{x\to 3}(x + 2) = 5}$$

$x$is the independent variable.

- 3 is the value that x approaches.
- 5 is the limiting value.

2. Meaning of Limit Notation

The general notation of a limit is:

$$\lim_{x\to a} f(x) = L$$

"The limit of $f(x)$ as x approaches a is equal to L."

- x= independent variable.
- a= value approached by x.
- $f(x)$= given function.
- L= limiting value.

Techtogreen

Types of Limits

A. Left-Hand Limit (LHL)

When x approaches a particular value from the left side, it is called the Left-Hand Limit.

It is represented by:

$$\lim_{x \to a^-} f(x)$$

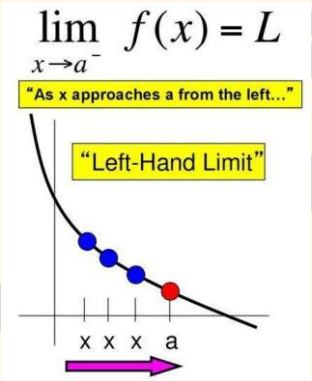
The negative sign indicates that x approaches a from values **smaller than a**.

Example:

Suppose x approaches 3 from the left.

$$x = 2.9, 2.99, 2.999$$

All these values are less than 3.



B. Right-Hand Limit (RHL)

When x approaches a particular value from the right side, it is called the Right-Hand Limit.

It is represented by:

$$\lim_{x \to a^+} f(x)$$

The positive sign indicates that x approaches a from values greater than a.

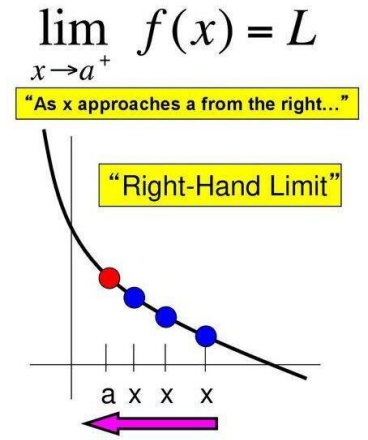
Example:

Suppose x approaches 3 from the right.

$$x = 3.1, 3.01, 3.001$$

All these values are greater than 3.

Technogreen



C. Two-Sided Limit

When x approaches a particular value from both the left and right sides, it is called a two-sided limit.

It is represented by:

$$\lim_{x \to a} f(x)$$

Important condition:

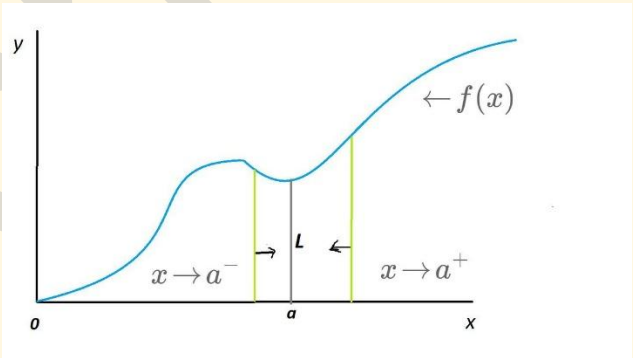
A two-sided limit exists only when:

$$\boxed{LHL = RHL}$$

That is,

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$$

If LHL and RHL are different, the limit does not exist.



Teotogreen

4. Basic Methods to Evaluate Limits

Method 1: Direct Substitution Method

In this method, directly substitute the value of x into the given function.

Note: If $x \to a$ and after putting $x = a$ in $f(x)$, we are not getting ∞ then follow this method.

Example 1:

Evaluate:

$$\lim_{x \to 2}(x^2 + 3x + 1)$$

Step 2: Substitute $x = 2$ in the function $x^2 + 3x + 1$.

$$= 2^2 + 3(2) + 1$$
$$= 4 + 6 + 1$$
$$\boxed{= 11}$$

Therefore,

$$\boxed{\lim_{x \to 2}(x^2 + 3x + 1) = 11}$$

Example 2:

Evaluate:

$$\lim_{x \to 1}(2x^3 + 5x - 4)$$

Substitute x = 1.

$$= 2(1)^3 + 5(1) - 4$$
$$= 2 + 5 - 4$$
$$\boxed{= 3}$$

Techtogreen

Method 2: Factorization Method

Sometimes direct substitution gives the indeterminate form:

$$\frac{0}{0}$$

In this situation, factorize the numerator and denominator and cancel the common factors.

Example 1:

Evaluate:

$$\lim_{x\to 2}\frac{x^2-4}{x-2},\ (a+b)(a-b)=a^2-b^2$$

Step 1: Substitute $x = 2$.

$$=\frac{2^2-4}{2-2}=\frac{0}{0}$$

This is an **indeterminate** form.

Step 2: Factorize the numerator.

We know that:

$$a^2-b^2=(a-b)(a+b)$$

Therefore,

$$=\lim_{x\to 2}\frac{(x-2)(x+2)}{x-2},\ \textbf{Since } x\to 2\textbf{, so } x\neq 2$$

Step 3: Cancel the common factor.

$$=\lim_{x\to 2}(x+2)$$

Step 4: Substitute $x = 2$.

$$=2+2$$

$$\boxed{=4}$$

Example 2:

Evaluate: $\lim_{x\to 3}\frac{x^2-9}{x-3}$

Step 1: Factorize.

$$=\lim_{x\to 3}\frac{(x-3)(x+3)}{x-3}$$

Step 2: Cancel the common factor.

$$=\lim_{x\to 3}(x+3)$$

Step 3: Substitute x = 3.

$$=3+3$$

$$\boxed{=6}$$

Techtogreen

Example 3:

Evaluate:

$$\lim_{x\to 4}\frac{\sqrt{x}-2}{x-4}$$

Step 1: Substitute x = 4.

$$=\frac{\sqrt{4}-2}{4-4}=\frac{0}{0}$$

Step 2: Multiply numerator and denominator by the conjugate.

$$=\lim_{x\to 4}\frac{(\sqrt{x}-2)(\sqrt{x}+2)}{(x-4)(\sqrt{x}+2)}$$

Step 3: Apply the identity.

$$=\lim_{x\to 4}\frac{x-4}{(x-4)(\sqrt{x}+2)}$$

Step 4: Cancel the common factor.

$$=\lim_{x\to 4}\frac{1}{\sqrt{x}+2}$$

Step 5: Substitute x = 4.

$$=\frac{1}{2+2}$$

$$\boxed{=\frac{1}{4}}$$

5. Important Standard Limits

Techtogreen

Students should remember the following standard limits.

No.	Standard limit	Answer
1	$\lim_{x\to 0} \frac{\sin x}{x}$	1
2	$\lim_{x\to 0} \frac{e^x - 1}{x}$	1
3	$\lim_{x\to 0} \frac{a^x - 1}{x}$	$\ln a$
4	$\lim_{x\to 0} \frac{\log(1+x)}{x}$	1

Note: Trigonometric standard limits are valid when angles are measured in radians.

Properties:

1. $\lim_{x\to a} \big(f(x) + g(x)\big) = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$, **provided both individual limits exist**
2. $\lim_{x\to a} \big(f(x) - g(x)\big) = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)$, **provided both individual limits exist**
3. $\lim_{x\to a} [f(x)g(x)] = \lim_{x\to a} f(x) \times \lim_{x\to a} g(x)$, **provided both individual limits exist**

Example

1. Evaluate:

$$\lim_{x\to 2}(x+3)(x+4)$$

Step 1:
$$= \lim_{x\to 2}(x+3) \times \lim_{x\to 2}(x+4)$$

Step 3: Calculate individual limits.

$$= (2+3) \times (2+4) = 5 \times 6 \boxed{= 30}$$

2. Evaluate:

$$\lim_{x\to 1}(x^2 + 2x)(3x+4)$$

Using the product rule,

$$= \lim_{x\to 1}(x^2 + 2x) \times \lim_{x\to 1}(3x+4)$$

Substitute $x = 1$.

$$= \big(1^2 + 2(1)\big) \times (3(1)+4) = (1+2) \times (3+4) = 3 \times 7 \boxed{= 21}$$

Techgreen

3. Evaluate: $\lim_{x\to 0}(x+2)\sin x$

Using the product rule,

$$= \lim_{x\to 0}(x+2) \times \lim_{x\to 0}\sin x$$

We know:

$$\lim_{x\to 0}(x+2) = 2$$

And,

$$\lim_{x\to 0}\sin x = 0$$

Therefore,

$$= 2 \times 0$$
$$\boxed{= 0}$$

4. Evaluate: $\lim_{x\to 0} x\left(\frac{\sin x}{x}\right)$

Using the product rule,

$$= \lim_{x\to 0} x \times \lim_{x\to 0}\frac{\sin x}{x}$$

We know:

$$\lim_{x\to 0} x = 0$$

And the standard limit:

$$\lim_{x\to 0}\frac{\sin x}{x} = 1$$

Therefore,

$$\lim_{x\to 0} x\left(\frac{\sin x}{x}\right) = 0 \times 1$$

5. Example:

$$\lim_{x\to 0}\frac{\tan x}{x} = \lim_{x\to 0}\frac{\frac{\sin x}{\cos x}}{x} = \lim_{x\to 0}\frac{\sin x}{x} \times \frac{1}{\cos x} = \lim_{x\to 0}\frac{\sin x}{x} \times \lim_{x\to 0}\frac{1}{\cos x} = 1 \times 1 = 1$$

6. Show that:

$$\lim_{x\to 0}\frac{1-\cos x}{x^2} = \frac{1}{2}$$

Indeterminate Forms

Sometimes direct substitution does not give a definite answer.
These are called indeterminate forms.
The seven commonly used forms are:

$$\boxed{\frac{0}{0},\ \frac{\infty}{\infty},\ 0 \times \infty}$$

$$\boxed{\infty - \infty,\ 0^0,\ 1^\infty,\ \infty^0}$$

Remember: An indeterminate form does not mean that the limit does not exist. It means that further simplification is necessary.

PART 2: CONTINUITY

Techtogreen